

Memo

The question I actually answered

Among SEC-reporting pharmaceutical and biotechnology firms, how does firm size, measured by annual revenue, relate to the distribution of clinical-trial development phases among industry-sponsored interventional trials led by these firms from 2016 to 2026?

The scope of the analysis was constrained by data availability and the cost of obtaining firm-level data. I therefore used the freely available AACT and SEC EDGAR databases, which narrowed the research question. To make the question measurable, I used the clinical trial phase as a proxy for the stage of drug development, with earlier phases representing earlier-stage development and later phases representing later-stage involvement.

Decision log / alternatives / what I chose not to do

Decision	Choice	Rationale / Trade-off
Trial-side data source	AACT	AACT was selected over Citeline and Cortellis. Although they offer broader drug-program and non-U.S. coverage, they are not free.
Trial type	Industry-sponsored interventional trials	The question concerns firms' active clinical development of drugs. Interventional trials directly capture a sponsor's involvement in testing an intervention. Observational studies do not represent the same development activity.
Study window	2016–2026	The original research question concerned the last decade, setting the analysis window to 2016–2026.
Firm-side data source	SEC EDGAR	SEC EDGAR was chosen because it provides standardized, publicly accessible annual financial data across firms and years. Alternatives such as PitchBook provide broader coverage of private companies, but cost money. The pipeline does not impute financial information for unmatched firms.
Definition of firm	Exact match to an SEC registrant	Sponsor names are normalized and matched, reducing the risk of incorrectly assigning a trial to the wrong company.

Decision	Choice	Rationale / Trade-off
Industry definition	SIC 2833, 2834, 2836, 8731	The SIC filter was added because the initial SEC-matched sponsor universe contained clearly unrelated companies; it is not a size classifier. The trade-off is that developers operating under another SIC code may be excluded.
Firm-size measure	Annual revenue	Revenue was selected as a consistently available measure of firm size over alternatives such as headcount, market capitalization, R&D spending, or hand-built lists. Firms between the thresholds are classified as OTHER. The thresholds were informed by IQVIA's <i>Emerging Biopharma's Contribution to Innovation</i> , reducing reliance on an arbitrary firm roster.
Time-varying size classification	Revenue in trial-start year	Firm classification is based on revenue in the trial's start year to account for changes in firm size over time. Revenue was not inflation-adjusted due to time constraints.
Phase number	Phase 1/2 conversion, Phase 4	Combination designations are reduced to a single phase, with the earlier phase selected to create one consistent phase measure per trial. Phase 4 is retained to capture the full Phase 1–4 distribution, though this may introduce bias because Phase 4 trials occur only after regulatory approval.
Trial attribution	Lead sponsor	For the primary exhibit, a multi-sponsor trial is attributed to the lead sponsor. This provides a consistent firm-level attribution rule for comparing clinical-development portfolios.
Primary comparison	Phase distribution by firm group	The final analysis compares the percentage of lead-sponsor trials in each phase within Big Pharma and Biotech. This directly addresses whether smaller and larger firms have different clinical-development-stage portfolios.

Further elaborated in Scratch.docx

Twenty more hours

biotech / is_pharma classification. Validate SIC-based classification against an external biotech index, and test R&D spending/intensity as an alternative way to distinguish biotech from pharma.

Therapeutic indication. Add therapeutic area to test whether differences in trial-phase distributions are driven by indication rather than firm size. Investigate further.

Inflation-adjusted revenue. Convert revenue to constant dollars and reapply the \$500M / \$10B thresholds to test whether inflation changes firm-size classifications.

Time effects / changing firm composition. Exclude recent trial cohorts or use a lagged end date to test whether limited follow-up makes newer trials appear disproportionately early-stage.

Code readability. I also know the code is not written as cleanly or efficiently as it could be. I combined several separate files and rerouted paths to consolidate everything into one script, so parts of it are a bit messy. I did review the logic, making sure the steps were consistent.

The one question I most wish I'd asked

Should the clinical-development pipeline account for where a drug originated, such as academic discovery? Should VC-backed biotech development, or internally developed pharmaceutical programs, be treated differently when assessing development stage and risk?

What I'm least confident about

Of 75,293 industry trials, only 5,347 (7.1%) are ultimately classified as Big Pharma or Biotech. Importantly, the missing data are not random: private and non-U.S. firms are less likely to have SEC data and may be smaller and earlier-stage, potentially introducing selection bias and limiting generalizability.

Missing Value in Raw Data

```
nct_id|nctm_download_date|description|study_first_submitted_date|results_first_submitted_date|disposition_first_submitted_date|last_update_submitted_date|study_fi
NCT03744910|2018-11-06|2025-04-03|2025-07-22|2018-11-13|2018-11-19|ACTUAL|2025-07-22|2025-07-23|ACTUAL|||2025-07-22|2025-07-23|ACTUAL|2019-10-14|ACTUAL|2019-1
NCT07628790|2026-06-01||2026-07-27|2026-06-01|2026-06-05|ACTUAL|||||2026-07-27|2026-07-28|ACTUAL|2026-06-01|ACTUAL|2026-06-01|2026-07|2026-07-31|2026-07-27|A
NCT04237545|2020-01-17||2026-07-24|2020-01-17|2020-01-23|ACTUAL|||||2026-07-24|2026-07-28|ACTUAL|2019-10-01|ACTUAL|2019-10-01|2026-07|2026-07-31|2026-07-01|A
NCT02327884|2014-12-30||2026-07-25|2014-12-30|2014-12-31|ESTIMATED|||||2026-07-25|2026-07-28|ACTUAL|2015-04-03|ACTUAL|2015-04-03|2026-05-06|2026-05-06|2022-0
NCT00359684|2006-08-01||2026-07-25|2006-08-01|2006-08-02|ESTIMATED|||||2026-07-25|2026-07-28|ACTUAL|1979-01-04|ACTUAL|1979-01-04|2026-04-13|2026-04-13|||||
NCT03412045|2017-12-05|2026-04-14|2026-06-30|2018-01-25|2018-01-26|ACTUAL|2026-06-30|2026-07-28|ACTUAL|||2026-06-30|2026-07-28|ACTUAL|2022-09-01|ACTUAL|2022-0
NCT05440617|2022-06-30||2026-07-25|2022-06-30|2022-07-01|ACTUAL|||||2026-07-25|2026-07-28|ACTUAL|2022-07-22|ACTUAL|2022-07-22|2026-06-03|2026-06-03|2041-09-2
NCT07633288|2026-06-02||2026-07-27|2026-06-02|2026-06-08|ACTUAL|||||2026-07-27|2026-07-28|ACTUAL|2026-05-01|ACTUAL|2026-05-01|2026-07|2026-07-31|2027-08-01|E
NCT03407053|2018-01-20|2021-05-18||2021-06-08|2018-01-20|2018-01-23|ACTUAL|2021-06-08|2021-06-29|ACTUAL|||||2021-06-08|2021-06-29|ACTUAL|2018-03-01|ACTUAL|2018-0
NCT05696483|2022-12-13||2026-07-27|2023-01-23|2023-01-25|ACTUAL|||||2026-07-27|2026-07-28|ACTUAL|2023-01-20|ACTUAL|2023-01-20|2026-07|2026-07-31|2026-09-01|E
NCT01027910|2009-12-08|2016-03-09|2025-07-13|2009-12-08|2009-12-09|ESTIMATED|2016-12-21|2017-02-14|ACTUAL|||2025-07-13|2025-07-24|ACTUAL|2009-02|2009-02-28|2
NCT05283356|2022-03-01||2026-07-27|2022-03-11|2022-03-16|ACTUAL|||||2026-07-27|2026-07-28|ACTUAL|2022-01-21|ACTUAL|2022-01-21|2026-07|2026-07-31|2028-06-30|E
NCT03224403|2017-07-07|2021-03-22|2019-04-02|2021-03-22|2017-07-18|2017-07-21|ACTUAL|2021-03-22|2021-04-15|ACTUAL|2019-04-02|2019-04-09|ACTUAL|2021-03-22|2021-0
NCT04153175|2019-11-03||2024-03-11|2019-11-05|2019-11-06|ACTUAL|||||2024-03-11|2024-03-13|ACTUAL|2020-08-21|ACTUAL|2020-08-21|2024-03|2024-03-31|2024-03-01|A
NCT04234529|2019-12-09||2021-12-08|2020-01-15|2020-01-21|ACTUAL|||||2021-12-08|2021-12-09|ACTUAL|2020-02-01|ACTUAL|2020-02-01|2021-12|2021-12-31|2021-06-11|A
NCT06464380|2024-06-13||2024-09-19|2024-06-13|2024-06-18|ACTUAL|||||2024-09-19|2024-09-23|ACTUAL|2024-10-30|ESTIMATED|2024-10-30|2024-09|2024-09-30|2025-10-0
NCT00346281|2006-06-27||2006-06-28|2006-06-28|2006-06-29|ESTIMATED|||||2006-06-28|2006-06-29|ESTIMATED|2006-06||2006-06-30|2006-06|2006-06-30|2008-06||2008-0
NCT0551923|2022-09-13||2022-09-21|2022-09-21|2022-09-23|ACTUAL|||||2022-09-21|2022-09-23|ACTUAL|2020-08-14|ACTUAL|2020-08-14|2022-09|2022-09-30|2025-04|ESTI
NCT05315115|2022-03-30||2024-10-23|2022-04-06|2022-04-07|ACTUAL|||||2024-10-23|2024-10-26|ACTUAL|2022-04-07|ACTUAL|2022-04-07|2024-10|2024-10-31|2022-12-01|A
NCT02908685|2016-09-19|2020-08-27||2024-04-01|2016-09-19|2016-09-21|ESTIMATED|2021-05-20|2021-06-15|ACTUAL|||2024-04-01|2024-04-24|ACTUAL|2016-10-19|ACTUAL|201
NCT02835274|2016-06-24|2019-12-18||2020-11-17|2016-07-13|2016-07-18|ESTIMATED|2020-01-28|2020-02-07|ACTUAL|||2020-11-17|2020-11-19|ACTUAL|2016-07-19|ACTUAL|201
NCT04327362|2020-03-25||2024-10-22|2020-03-26|2020-03-31|ACTUAL|||||2024-10-22|2024-10-24|ACTUAL|2021-10-01|ACTUAL|2021-10-01|2024-10|2024-10-31|2022-05-01|A
NCT02049489|2014-01-28||2018-08-09|2014-01-28|2014-01-30|ESTIMATED|||||2018-08-09|2018-08-13|ACTUAL|2013-12||2013-12-31|2018-08|2018-08-31|2017-03|ACTUAL|201
NCT04769323|2021-02-21||2025-03-23|2021-02-23|2021-02-24|ACTUAL|||||2025-03-23|2025-03-26|ACTUAL|2021-02-21|ACTUAL|2021-02-21|2025-03|2025-03-31|2023-06-15|A
NCT05368259|2022-05-06||2024-10-23|2022-05-06|2022-05-10|ACTUAL|||||2024-10-23|2024-10-26|ACTUAL|2022-05-12|ACTUAL|2022-05-12|2024-10|2024-10-31|2025-03-01|E
NCT06644755|2024-10-04||2025-07-21|2024-10-14|2024-10-16|ACTUAL|||||2025-07-21|2025-07-24|ACTUAL|2024-11-14|ACTUAL|2024-11-14|2025-07|2025-07-31|2028-11-30|E
NCT06315413|2024-03-10||2025-03-22|2024-03-14|2024-03-18|ACTUAL|||||2025-03-22|2025-03-26|ACTUAL|2025-05|ESTIMATED|2025-05-31|2024-08|2024-08-31|2025-10|ESTI
NCT04750928|2021-02-10||2026-07-25|2021-02-10|2021-02-11|ACTUAL|||||2026-07-25|2026-07-28|ACTUAL|2021-11-29|ACTUAL|2021-11-29|2026-06-03|2026-06-03|2029-12-3
```

Figure 1: Raw pipe-delimited records from data/AACT/studies.txt. Consecutive delimiters show fields with missing values.

File: data/AACT/studies.txt

Issue: Missing values.